

🔥 XAI: Interpreting GAN Latent Spaces

FAMAF - UNC

First Semester 2026



eXplainable
Artificial Intelligence

Disclaimer ---

This course is based on

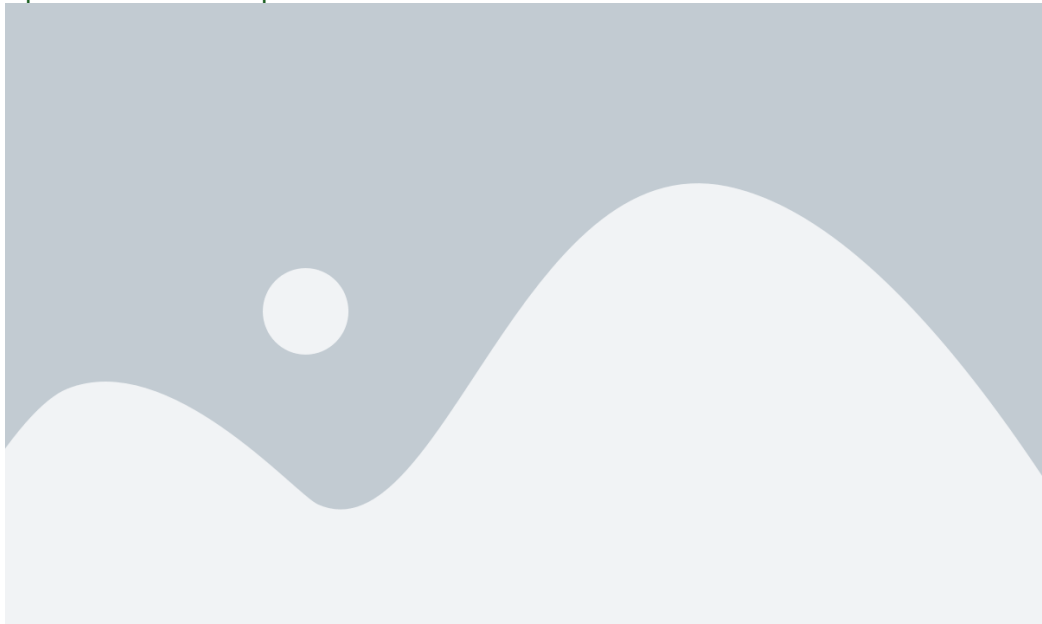
Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

<https://interpretable-ml-class.github.io/>

🍷 Paper 1: GANSpace ---



🍷 GANSpace: Motivation ---



Main Contributions ---

Core Idea

Apply PCA to GAN intermediate activations to discover interpretable editing directions in latent space — **no supervision required**.

- Works for both **StyleGAN** (**w**-space) and **BigGAN** (intermediate layers)
- StyleGAN editing: $\mathbf{w}' = \mathbf{w} + \alpha \mathbf{v}_k$
- BigGAN editing: $\mathbf{z}' = \mathbf{z} + \mathbf{U}\mathbf{x}$
- Controls correspond to semantic attributes: pose, lighting, color, shape

🍷 Background: Generative Adversarial Networks ---

- **Generator** G : maps latent z to image $G(z)$
- **Discriminator** D : distinguishes real vs. generated images
- Trained adversarially — generator improves to fool discriminator



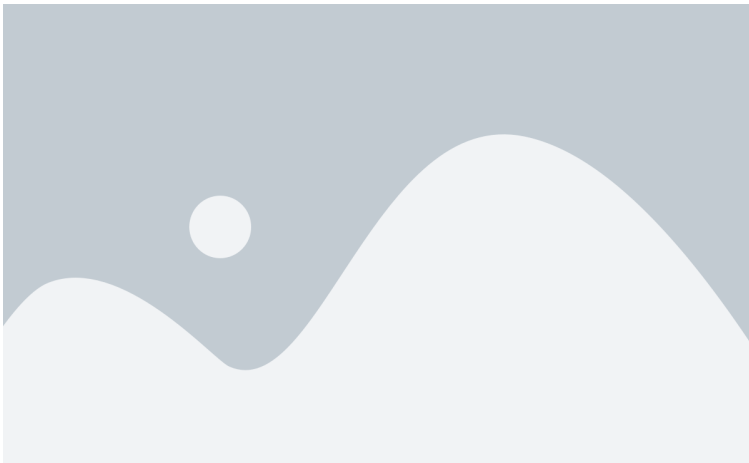
🍷 Background: BigGAN ---

- Class-conditional GAN for high-resolution ImageNet images
- Latent vector $\mathbf{z}$ is passed as **additional input** to all layers (no separate style vector)
- State-of-the-art for diverse, high-quality image generation



🍷 Background: StyleGAN ---

- Style-based generator: separates high-level attributes from stochastic variation
- Maps $\mathbf{z} \rightarrow \mathbf{w}$ via mapping network; $\mathbf{w}$ controls synthesis via AdaIN
- **w-space** is more disentangled than $\mathbf{z}$ -space



🍷 Background: PCA ---

- **Principal Component Analysis:** finds orthogonal axes of maximum variance
- Given data $\mathbf{y}_{1:N}$, PCA yields basis $\mathbf{V}$ and mean $\boldsymbol{\mu}$
- Project data: $\mathbf{x}_{1:N} = \mathbf{V}^T(\mathbf{y}_{1:N} - \boldsymbol{\mu})$
- First PCs capture the most variance; later PCs capture finer details

🍷 Related Work ---

- **Supervised directions:** require attribute labels to train classifiers in latent space
- **GAN analysis:** SeFa, closed-form factorization of generator weights
- GANSpace is **unsupervised** — no attribute annotations needed
- Related applications: camera motion, scene synthesis, image memorability

🍷 Methods: StyleGAN Approach ---

Idea: PCA directly on $\mathbf{w}$ -space samples

- 1 Sample N random $\mathbf{z}_{1:N}$, pass through mapping network to get $\mathbf{w}_{1:N}$
- 2 Apply PCA to $\mathbf{w}_{1:N} \rightarrow$ basis $\mathbf{V}$, mean μ
- 3 Represent each $\mathbf{w}$ in new basis: $\mathbf{x} = \mathbf{V}^T(\mathbf{w} - \mu)$

Editing

Move in the k -th principal direction with magnitude α :

$$\mathbf{w}' = \mathbf{w} + \alpha \mathbf{v}_k$$

Apply $\mathbf{w}'$ to some or all synthesis layers for global vs. localized edits.

🍷 Methods: BigGAN Approach ---

Issue: BigGAN has no separate style vector; $\mathbf{z}$ is passed as input at each layer



🍷 Methods: Granularity of Edits ---

Layerwise control

- **StyleGAN**: pass modified $\mathbf{w}'$ to some layers, original $\mathbf{w}$ to others
- **BigGAN**: pass modified $\mathbf{z}'$ to some layers
+ keep original $\mathbf{z}$ at start of network

Composed edits

- Edits can be *composed* by moving in **several** principal directions simultaneously
- Directions can be **labelled** by users to correspond to semantic concepts

🍷 Findings and Results ---



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🍲 Findings and Results ---



🍷 Analyzing GANs Through Principal Components ---

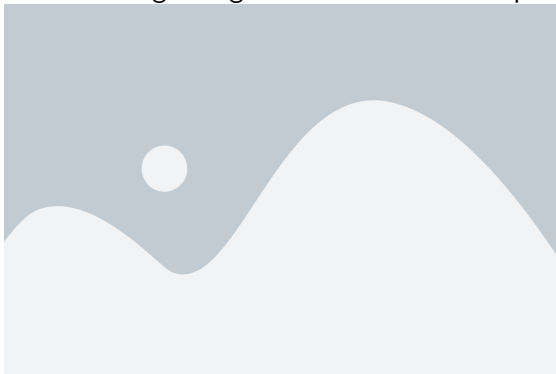
Limitations (inherited from dataset)

- StyleGAN faces cannot be translated in the image
- PCs for wrinkles/makeup have no effect on children/men



Entanglement/Superposition

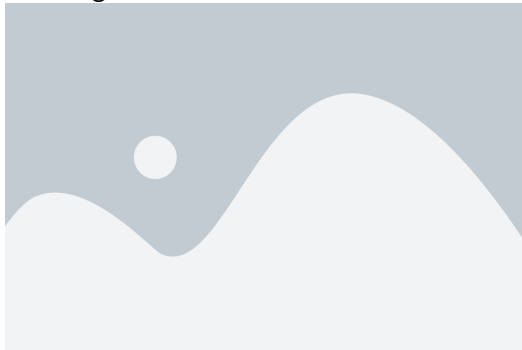
- One PC for StyleGAN cars corresponds to sportiness *and* open-road backgrounds
- Rotating a dog causes its mouth to open



🍷 Comparison with Related Techniques ---

GANSpace (PCA)

- Gives separated, ordered series of stylistic edit axes
- Random directions produce indiscernible changes



vs. Supervised methods

- Results similar to supervised approaches, without extra training
- Provides **many more** edit directions
- Some additional entanglement

🍷 Strengths / Weaknesses ---

Strengths

- Unsupervised and computationally simpler than supervised methods
- PC directions elucidate interpretable GAN latent representations
- After labelling directions, users can perform numerous edits on various styles

Weaknesses

- Approach limited to specific GAN architectures
- PCs are not *a priori* linked to semantic meaning
- Choice of which PCs to modify at which layers seems arbitrary
- Entanglement of PC styles hinders practicality
- Does this constitute actual **interpretability** for GANs?

🍷 GANSpace: Discussion Questions ---

- Are these principal components a useful tool for **interpreting** GANs?
- How does their utility compare with other interpretability methods we've studied?
- Most conceptually-aligned PCs for StyleGAN are in **w**-space; most useful PCs for BigGAN originate in the first linear layer (then projected to **z**-space)
- Do these findings provide novel insights into the models?
- How could this tool help **identify and reduce biases** in a model?

🍷 GANSpace: Summary ---

Experiment targets: **StyleGAN**, **BigGAN** (methods differed for the models)

- ① Changing the **first 20** PCs corresponds to changes in image **layout**, **configuration**, and **perspective**; later PCs dictate appearance and details
- ② StyleGAN's **first 100 PCs** sufficiently explain overall image appearance, distributed unimodally and almost independently
- ③ BigGAN's PCs seem to be **class-independent**

(Härkönen et al. 2020)

🍷 Paper 2: InterFaceGAN ---



InterFaceGAN: Roadmap ---

- Introduction
- Method
- Experiments

🍷 Background: What are GANs? ---

- **GAN**: generative models that generate realistic synthetic data mimicking training data
 - ▶ **Generator**: generates samples similar to training data
 - ▶ **Discriminator**: distinguishes generated (fake) from real samples
- **Latent representation**: a low-dimensional representation of an image
- **Latent space**: the set of all possible latent representations
 - ▶ Through adversarial training, the generator learns a mapping from latent space to real images



🍷 Background: GANs for Image Editing ---

- **Latent code:** the version of an image in the latent space
 - ▶ **GAN inversion:** identifying the latent code for a given image such that the generator could reconstruct it
- **Semantic editing:** editing the latent code to manipulate some features of the resulting image, such that only the desired features are changed





Related Work ---

- **Exploring latent spaces in GANs**

- ▶ Laine et al.: smoothly varying outputs across latent space
- ▶ Radford et al. observe the *vector arithmetic property*: `vector("King") - vector("Man") + vector("Woman") = vector("Queen")`
- ▶ Applications in camera motion, scene synthesis, memorability

- **Semantic face editing with GANs**

- ▶ Previous methods required carefully designed loss functions or specialized architectures
- ▶ No existing method to perform controlled facial editing by varying latent codes

Research Questions ---

- **How do semantics in the latent space originate, and how are they organized?**
 - ▶ Does there exist structure in the latent space relating to disentangled semantic representations?
- **What does a GAN actually learn with respect to the latent space?**
 - ▶ How does the GAN connect the latent space and the image semantic space?
- **How can the latent code be used for image editing?**
 - ▶ How are various semantic attributes of an individual's face (gender, age) determined and entangled?

🍷 Contributions and Main Findings ---

- Proposed **InterFaceGAN**, a framework to identify semantics encoded in the latent space of face synthesis models
 - ▶ GANs learn **latent subspaces** corresponding to specific attributes
- InterFaceGAN enables **semantic face editing with any pre-trained GAN**
 - ▶ Found theoretical and experimental results verifying that linear subspaces align with emerging semantics
- Applied InterFaceGAN to **real image editing**
 - ▶ Successfully edited attributes of real faces by varying the latent code

🍷 Method: Semantic Subspace ---

For any binary attribute, define the signed distance from hyperplane $\mathbf{n}$:

$$d(\mathbf{n}, \mathbf{z}) = \mathbf{n}^T \mathbf{z}$$

Key Hypothesis

The generator g satisfies:

$$f(g(\mathbf{z})) = \lambda d(\mathbf{n}, \mathbf{z})$$

where f scores the semantic attribute. The hyperplane $\mathbf{n}^T \mathbf{z} = 0$ separates the latent space into two semantic regions.

In practice, find $\mathbf{n}$ by training a **linear SVM** on labeled latent codes.

🍷 Method: Conditional Manipulation ---

To edit attribute A_1 while **preserving** attribute A_2 , project out the component of $\mathbf{n}_1$ along $\mathbf{n}_2$:

$$\mathbf{n}_1^* = \mathbf{n}_1 - (\mathbf{n}_1^T \mathbf{n}_2) \mathbf{n}_2$$



Move along $\mathbf{n}_1^*$ to change A_1 while remaining on the hyperplane of A_2 .

🍷 Experiment 1: Latent Space Separation ---

Model: PGGAN (Progressive Growing GAN) — trained on CelebA-HQ (30K images, 40 attributes)



🍷 Experiment 1: Hyperplane Assumption ---



Train 5 independent linear SVMs on pose, smile, age, gender, eyeglasses. Evaluate on validation set (6K samples) and full set (480K samples).

🍷 Experiment 1: Classification Accuracy ---



🍷 Experiment 1: Latent Space Visualization ---

Choose z far away from and on the decision boundary, then generate from it.



🍷 Experiment 2: Single Attribute Manipulation ---

Verify whether the semantics found by InterFaceGAN are manipulable.



🍷 Experiment 2: Distance Effect ---



🍷 Experiment 2: Artifacts Correction ---



Manually labelled 4K bad syntheses (with artifacts), trained a linear SVM to find the separation

🍷 Experiment 3: Conditional Manipulation ---

Study the disentanglement between different attributes.



🍷 Experiment 3: Conditional Manipulation Results ---



🍷 Experiment 4: Results on StyleGAN ---

StyleGAN extends PGGAN with:

Component	Description
Mapping Network	Casts $\mathbf{z}$ as $\mathbf{w}$ (“style vector”)
Synthesis Network	Uses style vector + noise via AdaIN
Bilinear Sampling	Replaces nearest-neighbor upsampling
Mixing Regularization	Uses 2 latent codes to generate



🍷 Experiment 4: W-Space vs Z-Space ---

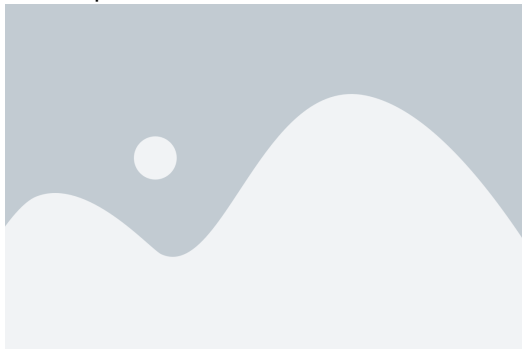


🍷 Experiment 5: Real Image Manipulation ---

Take a real face image $\rightarrow$ invert to latent code $\rightarrow$ use InterFaceGAN to edit.

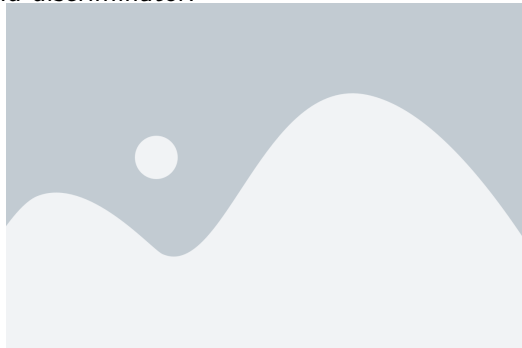
Latent Code Optimization

Optimize latent code with fixed generator to minimize pixel-wise reconstruction error.



Encoder-Based

Extra encoder learns inverse mapping from image to latent code by training with generator and discriminator.





InterFaceGAN: Discussion Questions ---

- 1 Do you believe the use of GANs for generating photo-realistic images should be **regulated**?
(Potential misuse)
- 2 Why do we think there are such simple linear boundaries separating binary semantics in the latent space (is it only for face GANs)?
- 3 How can analyzing semantic representations be used to **detect biases** in the training data?
- 4 Have you ever used GANs for photo generation, if so how realistic did you find them?
- 5 How relevant is this paper given that **GANs are progressively going out of style** (diffusion taking over)?

Thank You!

References --- |

Härkönen, Erik, Aaron Hertzmann, Jaakko Lehtinen, and Sylvain Paris. 2020. "GANSspace: Discovering Interpretable Gan Controls." In *Advances in Neural Information Processing Systems*.

Shen, Yujun, Jinjin Gu, Xiaoou Tang, and Bolei Zhou. 2020. "Interpreting the Latent Space of Gans for Semantic Face Editing." In *Proceedings of the Ieee/Cvf Conference on Computer Vision and Pattern Recognition*.